

From AI anxiety to strategic regulation: How university students transform generative AI into a strategic learning resource

Eunjeon Kim¹

Graduate School of Education, Dankook University, Republic of Korea

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ABSTRACT

This study examined how university students engaged with generative AI as a strategic learning resource in academic writing tasks, with particular attention to the role of AI anxiety. Using an explanatory sequential mixed-methods design, survey data and post-task written reflections were collected from 107 university students. The quantitative analysis examined the relationships among AI anxiety, evaluative capacity, ethical awareness, and regulatory engagement, while the qualitative analysis explored how students described verifying, revising, selectively adopting, or rejecting AI-generated content during actual academic writing tasks. The findings showed that AI anxiety was positively associated with verification and revision behaviors, whereas evaluative capacity and ethical awareness were stronger predictors of more active, selective, and reflective forms of AI-supported engagement. Qualitative findings further indicated that effective AI use depended not simply on willingness to use the tool, but on students' capacity to question outputs, revise selectively, and maintain authorship responsibility. These findings suggest that AI literacy in higher education may be understood less as technology acceptance and more as a form of regulatory competence grounded in evaluative judgment and ethical responsibility.

1. Introduction

Generative artificial intelligence (AI) has rapidly become embedded in higher education, reshaping how students access information, generate ideas, and complete academic tasks. Tools such as large language models now participate directly in processes of reading, writing, summarizing, and revising, thereby extending the scope of technological mediation in learning and blurring traditional boundaries between human cognition and automated knowledge production. In response to this shift, a growing body of research has sought to understand how students perceive and respond to generative AI, with particular attention to affective and attitudinal dimensions such as anxiety, attitudes, and acceptance.

Within this literature, prior studies have made substantial contributions by conceptualizing and empirically validating AI anxiety as a salient psychological construct in educational contexts. Large-scale survey studies have demonstrated robust associations between AI anxiety and students' motivation, attitudes toward AI, and intentions to engage in AI-related learning activities, often using predictive modeling techniques such as structural equation modeling (Wang et al., 2024; Wang & Wang, 2022). Parallel research has further examined general

attitudes toward AI and their associations with personality traits, trust, and demographic factors, clarifying individual differences in learners' receptivity to AI technologies (Schepman & Rodway, 2023; Kaya et al., 2024).

However, despite these contributions, existing research remains limited in its ability to explain how learning with generative AI is enacted. Much of the literature conceptualizes students primarily as respondents to AI, whose engagement is explained in terms of acceptance, resistance, or behavioral intention. This orientation reflects broader adoption-centered approaches that explain AI use primarily in terms of uptake, intention, and general engagement tendencies. While such approaches are useful for explaining why learners choose to use AI, they offer limited insight into what learners do with AI during academic work, particularly the moment-to-moment practices through which they interrogate, revise, resist, or appropriate AI-generated content.

Recent scholarship, particularly work published since 2020, suggests that students' engagement with generative AI is more complex and dynamic than adoption-oriented perspectives imply. In academic settings, learners often approach AI-generated content with uncertainty, caution, and ethical concern, especially when judging accuracy, authorship, bias, and academic integrity (Kaya et al., 2024; Ng et al., 2021; Wang et al.,

E-mail address: wingit@dankook.ac.kr.

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2024). Rather than simply accepting or rejecting AI outputs, students may check information, rewrite problematic content, or selectively incorporate useful suggestions into their work. These findings indicate that generative AI use is not merely a matter of uptake, but a process of ongoing judgment and regulation.

From this perspective, AI anxiety provides an important entry point for the present study. Although prior research has often treated anxiety primarily as a negative factor that reduces motivation or discourages engagement (Wang et al., 2024), educational psychology suggests that anxiety may also heighten attention and prompt greater evaluative control under conditions that require judgment and monitoring (Alpert & Haber, 1960; Piniel & Csizér, 2013). In the context of generative AI, anxiety may therefore function not only as a barrier to use but also as a productive signal that encourages learners to scrutinize outputs more carefully and avoid uncritical reliance. For this reason, examining AI anxiety offers a useful way of understanding how students move from simple AI use toward more deliberate regulatory engagement.

This possibility is especially important in writing-related academic tasks. When students use generative AI to support writing, they must do more than produce text efficiently. They must also determine whether AI-generated content is accurate, appropriate, ethically useable, and compatible with their own authorship. In such contexts, the educational issue is not simply whether students use AI, but how they regulate that use through verification, revision, selective uptake, and rejection.

To address this gap, the present study examines university students' regulatory engagement with generative AI in academic writing tasks, with particular attention to the role of AI anxiety. In this study, AI anxiety is treated as a directly measured affective construct, whereas strategic regulation is used as an analytic term to interpret how students manage AI use through checking, revising, selectively adopting, or rejecting AI-generated content in light of evaluative judgment and ethical awareness. Related concepts such as epistemic unease and evaluative vigilance are therefore treated as interpretive lenses rather than as standalone psychometric variables. Accordingly, the present study does not claim to have directly measured epistemic vigilance; rather, it uses the observed relationship between AI anxiety and regulatory engagement, together with students' reflective accounts, to interpret how such an epistemic stance may be manifested in practice. By combining survey data with students' reflective accounts, this study aims to clarify how AI anxiety is associated with regulatory engagement and how students describe using generative AI as a strategic learning resource.

1.1. Research questions

RQ1. How is AI anxiety associated with university students' regulatory engagement with generative AI in academic writing tasks?

RQ2. How do university students describe regulating their use of generative AI through verification, revision, selective uptake, and rejection during academic writing tasks?

RQ3. How do students' reflective accounts illuminate the roles of evaluative judgment and ethical awareness in strategic AI use?

2. Theoretical background

2.1. AI anxiety beyond technological fear: why generative AI changes the equation

AI anxiety has emerged as a central construct in recent discussions of human-AI interaction, particularly in educational contexts where learners increasingly engage with complex and opaque AI systems (Wang et al., 2024; Wang & Wang, 2022). Although rooted in earlier notions of computer anxiety and technophobia, which were typically associated with usability concerns and loss of technical control (Heinssen et al., 1987; Parasuraman, 2000), AI anxiety in generative AI environments may involve a broader form of concern related not only to

operating a system, but also to judging the credibility, appropriateness, and consequences of AI-generated content.

Unlike prior educational technologies, generative AI produces content that often appears fluent and authoritative, thereby complicating the distinction between retrieving information and generating knowledge claims. In educational settings, this may heighten learners' concern about accuracy, authorship, bias, and responsibility for what is incorporated into academic work (Johnson & Verdicchio, 2017; Ng et al., 2021; Selwyn, 2019). For this reason, AI anxiety in such contexts should not be understood only as a form of technical fear. It may also be related to learners' heightened caution when interpreting and using AI-generated output.

Importantly, this study distinguishes between directly measured constructs and interpretive concepts. AI anxiety is treated as a measured affective construct, whereas concepts such as epistemic unease and epistemic vigilance are used interpretively to explain why anxiety may, in some cases, coincide with closer scrutiny, greater hesitation, and more deliberate regulation of AI use during academic tasks. Accordingly, AI anxiety is considered here as a measured variable that may be associated with more cautious forms of engagement, rather than as a uniformly negative barrier to AI use. Table 1 summarizes this distinction.

2.2. Limits of attitude- and acceptance-centered models of AI use

Parallel to anxiety-focused research, a substantial literature has examined students' attitudes toward AI and their correlates. Studies using instruments such as the General Attitudes toward Artificial Intelligence Scale have shown systematic associations with personality traits, institutional trust, and demographic variables, clarifying individual differences in AI receptivity (Schepman & Rodway, 2023; Kaya et al., 2024). These contributions have been instrumental in explaining AI acceptance.

Nevertheless, attitude-based models remain limited in their capacity to explain learning. Grounded largely in adoption-centered frameworks such as the Technology Acceptance Model, this literature conceptualizes engagement as a decision to use technology based on perceived usefulness and ease of use (Davis, 1989; Venkatesh et al., 2003). While effective for predicting uptake, such models are less well suited to explaining how students work with AI output once it has been generated. In many cases, they treat use as a relatively stable decision outcome rather than as an ongoing process of judgment, revision, and selective uptake during task performance.

In generative AI contexts, where outputs may be fluent yet epistemically unstable, the central pedagogical issue is not whether students intend to use AI, but how they evaluate, adapt, and integrate AI-generated information into their own knowledge construction (Ng et al., 2021; Selwyn, 2019). This limitation is especially important in academic writing tasks, where learners must decide what can be accepted, what requires modification, and what should be rejected. Existing models therefore offer limited theoretical tools for capturing these processes, leaving a gap between dispositional predictors and enacted learning behavior.

2.3. Generative AI as an epistemic presence in learning

Recent scholarship increasingly conceptualizes generative AI not simply as a neutral instructional aid, but as a system whose outputs participate in meaning-making and influence how knowledge is evaluated (Johnson & Verdicchio, 2017; Ng et al., 2021; Selwyn, 2019). More recent studies on AI-supported academic work and AI literacy in higher education likewise suggest that students' engagement with generative AI involves not only access and efficiency, but also trust, evaluation, revision, and responsibility. Recent reviews in Computers & Education and Computers & Education: Artificial Intelligence further indicate that student engagement with generative AI is shaped by affective,

Table 1
Definitions of measured and interpretive concepts used in the study.

Concept	Definition	Status in this study	Role
AI anxiety	Students' reported concern, uncertainty, or discomfort regarding the use of generative AI in academic tasks	Directly measured construct	Affective predictor in the quantitative analysis
Evaluative capacity	Students' tendency to assess, verify, and revise AI-generated content	Directly measured construct	Predictor/interpretive support
Ethical awareness	Students' sensitivity to originality, responsibility, and appropriate use of AI-generated content	Directly measured construct	Predictor/interpretive support
Regulatory engagement	Students' reported tendency to verify, revise, selectively adopt, or reject AI-generated content during academic writing tasks	Directly measured construct	Outcome variable
Epistemic unease	A cautious interpretive stance toward the credibility or trustworthiness of AI-generated content	Interpretive concept	Used to explain cautious engagement with AI
Epistemic vigilance	Closer scrutiny and monitoring of AI-generated output during task completion	Interpretive concept	Used to interpret verification and monitoring
Ethical self-monitoring	Reflective attention to whether AI-supported work remains one's own and can be responsibly claimed	Interpretive concept	Used to interpret responsibility and ownership concerns

pedagogical, and institutional conditions, and that higher education is moving from questions of simple adoption toward questions of meaningful, responsible, and pedagogically guided use (Chiu, 2024; Lee et al., 2024; Lo et al., 2024). By producing authoritative-sounding text, generative AI challenges assumptions about authorship, expertise, and epistemic validation, creating learning conditions characterized by informational abundance and epistemic uncertainty.

Empirical studies indicate that learners frequently engage in verification, reinterpretation, selective appropriation, and rejection of AI outputs rather than uncritical acceptance (Ng et al., 2021; Wang et al., 2024). These practices reflect active negotiation of epistemic authority, yet they are rarely theorized explicitly within AI anxiety or attitude research. When acknowledged, regulation is often subsumed under broad constructs such as self-efficacy or critical thinking, obscuring the mechanisms through which learners manage epistemic risk and ethical responsibility.

For the purposes of the present study, these prior findings are important because they suggest that meaningful AI use in education is not exhausted by acceptance or frequency of use. Instead, students' engagement with AI may involve recurring acts of checking, revising, comparing, and withholding acceptance. This makes generative AI particularly relevant to discussions of evaluative judgment and responsible academic decision-making.

2.4. Strategic regulation as a framework for AI-supported learning

To address these limitations, this study advances strategic regulation as an analytic framework for understanding how students manage generative AI during academic task performance. In this study, strategic regulation does not refer to a standalone psychometric construct directly measured by a single scale. Rather, it is used as an interpretive term to describe learners' active management of AI use through verification, revision, selective uptake, and rejection in light of evaluative judgment and ethical awareness.

While informed by self-regulated learning theory (Pinel & Csizér, 2013; Zimmerman, 2000), this framework is adapted to the specific conditions of generative AI use, where the learner must regulate not only personal effort or strategy use, but also the status of externally generated content that may be fluent, persuasive, and uncertain in quality. Within this perspective, AI anxiety is not assumed to operate as a simple obstacle. Instead, it is treated as a possible affective condition associated with more careful monitoring and selective engagement, particularly when students perceive epistemic or ethical risk in AI-generated content.

Accordingly, the framework proposed here is intended to organize the relationship among three elements central to this study: AI anxiety as an affective predictor measured quantitatively; students' regulatory engagement with AI as reflected in practices such as checking, revising, and selective acceptance; and the interpretive roles of evaluative judgment and ethical awareness in explaining how students manage AI use in

context. In this sense, strategic regulation integrates affective, cognitive, and ethical dimensions of AI-supported learning without claiming that all these dimensions were measured in identical ways or at the same analytic level.

Fig. 1 presents the proposed framework of strategic regulation in AI-mediated learning.

3. Methodology

3.1. Research design and alignment with research questions

This study employed an explanatory sequential mixed-methods design to examine university students' regulatory engagement with generative artificial intelligence (AI) in academic writing-related contexts. A mixed-methods approach was appropriate because the study addressed both (a) the quantitative relationship between AI anxiety and regulatory engagement and (b) students' qualitative accounts of how they verified, revised, selectively adopted, or rejected AI-generated content during academic tasks. Whereas prior research on AI anxiety has relied primarily on large-scale surveys and predictive modeling to explain attitudes or intentions (Wang et al., 2024; Wang & Wang, 2022), such approaches offer limited insight into how students describe regulating AI use during actual task performance.

To address this gap, survey data were used together with students' reflective accounts, allowing AI use to be examined not only as an attitudinal phenomenon but also as a situated learning practice. In the present study, the three research questions were aligned with distinct but complementary analytic components: (a) a quantitative examination of the association between AI anxiety and regulatory engagement (RQ1), (b) a qualitative analysis of students' descriptions of verification, revision, selective uptake, and rejection during AI-supported writing tasks (RQ2), and (c) an interpretive integration of qualitative and quantitative findings to clarify the roles of evaluative judgment and ethical awareness in strategic AI use (RQ3). This alignment was intended to ensure consistency between the research questions, data sources, and analytic procedures, while avoiding the treatment of interpretive concepts as directly measured standalone variables.

Table 2 presents the alignment among the research questions, data sources, and analytic approaches used in the study.

3.2. Participants and learning context

Participants were 107 undergraduate students enrolled in a general education English course at a university where generative AI tools were not formally embedded in the curriculum but were readily accessible. The sample reflects a bounded instructional context and is consistent with mixed-methods research that prioritizes analytic generalization and theory refinement over population-level inference (Creswell & Plano Clark, 2018). More specifically, the study was situated in a

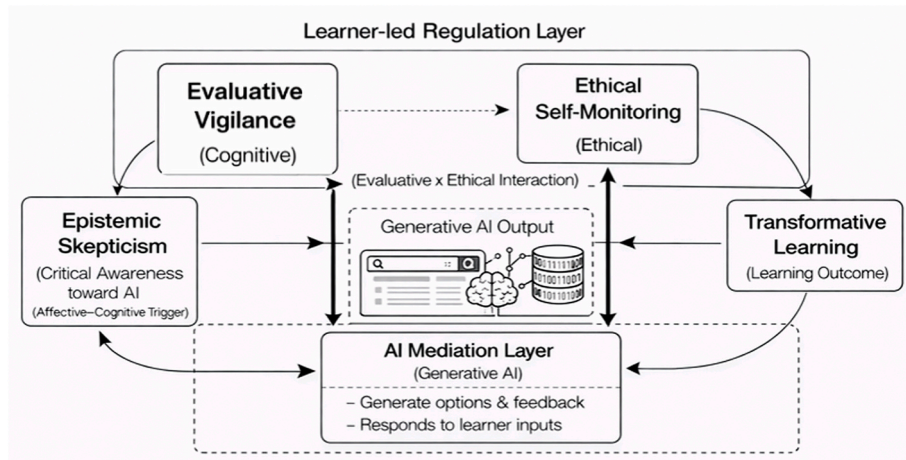


Fig. 1. Dual-layer conceptual framework of strategic regulation in AI-mediated learning.

Note. The figure presents the proposed framework as a heuristic representation rather than a validated causal model. It illustrates the conceptual relationships among AI anxiety and several higher-order interpretive concepts used in the study, including epistemic unease, evaluative vigilance, ethical self-monitoring, and more reflective AI-supported learning. Not all elements shown in the figure were directly measured as standalone variables; some are included as interpretive concepts used to explain students' regulatory engagement with generative AI.

Table 2
Alignment of research questions, data sources, and analytic approaches.

Research Question	Data Source	Analytic Focus	Analytic Approach
RQ1.	Survey data	Association between AI anxiety and regulatory engagement	Quantitative analysis
RQ2.	Reflective accounts	Enacted regulatory practices during AI-supported writing	Qualitative content analysis
RQ3.	Reflective accounts and integrated interpretation of survey results	Interpretive roles of evaluative judgment and ethical awareness in strategic AI use	Mixed-methods interpretive integration

South Korean university English-writing context in which students used generative AI to support summarizing, drafting, and revising short academic texts. The findings should therefore be interpreted in relation to this institutional, curricular, linguistic, and task-based setting rather than as directly generalizable to all higher-education contexts.

This participant group was selected to capture authentic patterns of student-initiated AI use in routine academic writing-related tasks. Rather than examining instructor-mandated integration, the study focused on how students voluntarily engaged with generative AI and how they described their own regulatory decisions during task completion. Because the study was conducted in a general education English course with relatively short writing tasks, the emphasis was on students' evaluative and regulatory practices in context rather than on broad claims about AI use across disciplines, genres, or institutional settings.

To enhance reproducibility, the AI-use episode was defined as a single task cycle in which students (a) selected one academic writing-related task (e.g., summarization of a short reading, drafting a short argumentative paragraph, or revising an existing draft) (b) consulted a generative AI tool (e.g., ChatGPT or comparable writing assistants), via their personal devices, and (c) submitted a brief written reflection immediately after task completion. Reflections were elicited using a standardized prompt asking students to report (i) what they asked the AI, (ii) what they accepted, modified, or rejected, and (iii) why (e.g., credibility, coherence, originality, and authorship concerns). Thus, the qualitative data consisted of short post-task written reflections produced immediately after a single AI-supported writing activity. These prompts

were designed to elicit regulatory actions rather than simple frequency of use, thereby supporting the qualitative analysis of verification, revision, selective uptake, and rejection described in RQ2. Because AI access was student-initiated, the specific tool version and interface could vary; therefore, the study focuses on learners' regulatory actions during AI-mediated writing rather than tool-specific performance.

All participants reported prior exposure to generative AI tools. Informed consent was obtained, and all data were anonymized prior to analysis. In accordance with institutional policies governing routine classroom-based educational activities, formal ethics committee approval was not required.

Descriptive information about participants' gender, academic major, year of study, and prior AI experience is summarized in Table 3. The sample was predominantly composed of first-year undergraduates, reflecting the structure of the general education curriculum in which the study was conducted. Despite this distribution, participants represented both humanities and STEM backgrounds and displayed varied levels of familiarity with generative AI tools, providing sufficient diversity for examining regulatory engagement patterns.

3.3. Data sources

3.3.1. Survey measures

Quantitative data were collected through a structured questionnaire measuring AI anxiety, evaluative capacity, ethical awareness, and regulatory engagement in AI-supported academic writing tasks. To respond to concerns about construct clarity, the survey focused only on variables

Table 3
Demographic characteristics and AI exposure level ($N = 107$).

Variable	Category	<i>n</i>	%
Gender	Female	58	54.2
	Male	49	45.8
Academic Major	Humanities & Social Sciences	61	57.0
	STEM (Science, Technology, Engineering, Mathematics)	46	43.0
Year of Study	First Year	102	95.3
	Second Year	3	2.8
	Third Year	2	1.9
Prior AI Experience	No direct use before course	18	16.8
	Occasional use (ChatGPT, AI writing tools)	59	55.1
	Frequent use ($\geq$ weekly)	30	28.0

Note. Percentages may not total 100 due to rounding.

that were directly measured in the present study; interpretive concepts used later in the discussion (e.g., epistemic unease or evaluative vigilance) were not treated as standalone survey constructs.

AI anxiety was assessed using a validated multidimensional scale encompassing both learning-related and sociotechnical concerns (Wang & Wang, 2022). In the present study, AI anxiety refers to students' reported concern, uncertainty, or discomfort regarding the use of generative AI in academic tasks. Evaluative capacity captured students' tendencies to assess, verify, and revise AI-generated content, while ethical awareness reflected their sensitivity to authorship, academic integrity, and responsible AI use (Ng et al., 2021). Both constructs were measured through multi-item Likert-type scales adapted from prior research on AI literacy and academic integrity (Ng et al., 2021). Each subscale included five items rated on a 5-point scale (1 = strongly disagree, 5 = strongly agree). For example, AI anxiety items addressed concern or discomfort regarding AI use in academic tasks, whereas evaluative capacity items focused on checking and revising AI-generated content. Full item examples are provided in Appendix A, and composite scores for each construct were calculated by averaging the responses across the five items in each subscale. Cronbach's α coefficients for the four measured constructs ranged from .79 to .86, indicating acceptable internal consistency.

Regulatory engagement, the focal outcome variable in the quantitative analysis, was operationalized as students' reported tendencies to verify, revise, selectively adopt, or reject AI-generated content during academic writing tasks rather than as learning intentions alone. This operationalization was chosen to align the quantitative component with the study's emphasis on enacted regulation rather than technology acceptance or behavioral intention. For clarity and consistency with the revised research questions, this construct is referred to throughout the manuscript as regulatory engagement.

Table 4 summarizes the directly measured constructs, their analytic roles, scale characteristics, illustrative item content, and reliability estimates. As shown in the table, all four constructs were operationalized through five-item Likert-type subscales, and each composite variable was calculated as the mean of its constituent items.

Taken together, the survey data captured the measured affective, evaluative, ethical, and behavioral dimensions of students' engagement with generative AI. This quantitative component was designed to clarify whether AI anxiety was associated with regulatory engagement and how evaluative capacity and ethical awareness contributed to that relationship.

3.3.2. Reflective task accounts

Qualitative data consisted of students' written reflections on AI-assisted academic tasks. Prompts elicited descriptions of AI use, experienced uncertainties, and decisions to evaluate, modify, or reject AI-generated content. These reflections provided insight into learners' regulatory reasoning and ethical deliberation and complemented survey data by capturing the processual and enacted dimension of strategic regulation (Piniel & Csizér, 2013).

More specifically, the reflective prompts were designed to elicit

accounts of what students asked the AI to do, what parts of the output they accepted, modified, or rejected, and why they made those decisions. In this sense, the prompt structure explicitly targeted students' post-task explanations of regulatory action rather than general attitudes toward AI. This structure was intended to generate evidence relevant to RQ2 by making students' verification, revision, selective uptake, and rejection practices visible in their own words. Accordingly, the qualitative data were used not to measure additional latent constructs, but to illuminate how students described their regulatory engagement in context.

This qualitative component therefore represented the experiential and interpretive layer of the research design. Rather than treating epistemic unease, evaluative vigilance, or ethical self-monitoring as directly measured variables, the analysis used students' reflective accounts to explore whether such orientations could be interpreted in their descriptions of AI-mediated academic work.

3.4. Data analysis

3.4.1. Quantitative analysis

Quantitative data were analyzed using descriptive statistics and hierarchical regression analyses to examine relationships among AI anxiety, evaluative capacity, ethical awareness, and regulatory engagement. Prior to analysis, assumptions of normality, linearity, independence, and homoscedasticity were verified. To ensure the validity of the regression models, multicollinearity diagnostics were conducted by inspecting tolerance and variance inflation factor (VIF) values; all VIFs were below 3.0, indicating that multicollinearity was not a concern. Statistical significance was determined at $\alpha = .05$, and effect sizes were estimated using Cohen's f^2 to complement significance testing (with thresholds of .02, .15, and .35 representing small, medium, and large effects, respectively; Cohen, 1988). These diagnostic procedures ensured that the regression results reflected genuine relationships among predictors rather than artifacts of variable interdependence.

This quantitative analysis primarily addressed RQ1 by examining whether AI anxiety was associated with students' reported regulatory engagement during AI-supported academic writing tasks. Evaluative capacity and ethical awareness were included as additional predictors because they were central to the study's conceptual framing and allowed the analysis to examine whether the relationship between AI anxiety and regulatory engagement remained meaningful when evaluative and ethical dimensions were considered simultaneously.

In addition to hierarchical regression, a moderated mediation analysis was conducted using Hayes' PROCESS macro (Model 7; Hayes, 2018) to examine whether ethical awareness moderated the effect of AI anxiety on evaluative capacity (i.e., the first-stage path), and whether AI anxiety was indirectly associated with regulatory engagement via evaluative capacity conditional on ethical awareness. All variables were mean-centered prior to analysis, and 5000 bootstrap samples were used to estimate bias-corrected confidence intervals for conditional indirect effects.

Because the revised framing of the study treats several higher-order

Table 4
Overview of directly measured survey constructs.

Construct	Analytic role	No. of items	Response scale	Example focus	Reliability
AI anxiety	Predictor	5	5-point Likert	Concern, uncertainty, and discomfort regarding AI use in academic tasks	.84
Evaluative capacity	Predictor/interpretive support	5	5-point Likert	Assessing, checking, revising, and judging AI-generated content	.81
Ethical awareness	Predictor/interpretive support	5	5-point Likert	Authorship, academic integrity, responsible AI use	.86
Regulatory engagement	Outcome	5	5-point Likert	Verification, revision, selective uptake, and rejection during AI-assisted writing	.79

Note. The table includes only constructs directly measured through the survey. Interpretive concepts discussed later in the paper were not operationalized as independent psychometric variables.

concepts as interpretive rather than directly measured constructs, the results of the PROCESS analysis are presented cautiously as evidence of conditional statistical relationships among measured variables, not as definitive confirmation of the full conceptual framework shown in Fig. 1.

3.4.2. Qualitative analysis

Qualitative data were analyzed using qualitative content analysis to identify patterns of regulatory engagement in students' reflections. The analysis proceeded through three iterative rounds of coding: first, open coding of regulatory actions; second, axial coding linking those actions to evaluative and ethical judgments; and third, selective coding to construct higher-order regulatory patterns. To increase analytic transparency, the coding scheme was organized around three dimensions: (a) regulatory actions (e.g., verification, revision, selective uptake, rejection), (b) evaluative reasoning, and (c) ethical considerations. Two independent coders analyzed all qualitative data using this coding scheme. The coding categories were developed in relation to the research questions, the reflective prompts, prior literature on self-regulated learning and AI-supported learning (e.g., Ng et al., 2021; Piniel & Csizér, 2013; Zimmerman, 2000), and recurring patterns identified in the dataset. Because the study adopted an explanatory sequential mixed-methods design, the qualitative analysis was used to interpret and elaborate the quantitative findings rather than to treat all higher-order concepts as directly measured variables. Inter-coder reliability, calculated using Cohen's κ , exceeded .70 across the major coding categories, indicating substantial agreement (Landis & Koch, 1977). Any disagreements in coding were resolved through discussion and consensus between the two coders. Representative excerpts were then selected to ensure analytic transparency.

This analysis primarily addressed RQ2 by examining how students described regulating their use of generative AI in practice. It also informed RQ3 by identifying how evaluative judgment and ethical awareness could be interpreted in students' written accounts. Importantly, the qualitative analysis was not used to claim that all theoretical concepts in the study were directly observed as discrete variables; rather, it was used to interpret how students' descriptions reflected recurring patterns of caution, judgment, revision, and responsibility during AI-assisted tasks.

3.5. Integration of quantitative and qualitative findings

Integration occurred at the interpretation stage through a joint display approach (Creswell & Plano Clark, 2018). Quantitative analyses identified relational patterns among the directly measured affective (AI anxiety), cognitive (evaluative capacity), and ethical (ethical awareness) variables, while qualitative findings illustrated how these associations were enacted in learners' regulatory practices during AI-assisted tasks. This integrative approach strengthened explanatory depth by linking statistical associations to observable regulatory behaviors and enabled a more cautious interpretive account of how affective, evaluative, and ethical dimensions were related to students' regulatory engagement.

Table 5 presents a joint display that integrates the statistical results with representative qualitative excerpts, illustrating how key variables were reflected in practice. This approach enabled direct comparison across datasets and strengthened inference by demonstrating how quantitative relationships corresponded to observed behavioral and reflective evidence, while maintaining a distinction between directly measured survey constructs and interpretive qualitative insights.

For example, the quantitative finding that higher AI anxiety predicted greater verification behavior ($\beta = .24, p < .01$) was supported by qualitative accounts describing learners' vigilant questioning of AI outputs ("I checked two other sources before trusting the AI's answer"). Similarly, the interaction between evaluative capacity and ethical awareness, which was identified in the hierarchical regression

Table 5
Joint display of quantitative and qualitative findings (after Creswell & Plano Clark, 2018).

Quantitative Finding	Qualitative Theme/Excerpt	Interpretive Contribution
Higher AI anxiety positively predicted verification ($\beta = .24, p < .01$).	"Because I was worried about plagiarism, I kept questioning every sentence the AI gave me."	Suggests that concern about AI use may be associated with closer scrutiny of AI-generated content.
Evaluative capacity predicted revision and selective integration ($\beta = .46, p < .001$).	"I rewrote most of the text, only keeping the logical structure the AI suggested."	Indicates that stronger evaluative capacity may support active revision and selective uptake rather than passive acceptance.
Interaction between evaluative capacity $\times$ ethical awareness was associated with higher levels of regulatory engagement ($\beta = .41, p < .01$).	"AI helped me think, but the final idea felt more mine because I changed it so much."	Suggests that ethical awareness may strengthen how students translate evaluation into more active and responsible AI use.

interaction analysis () as predicting higher levels of regulatory engagement, was reflected in students' reflections describing stronger responsibility and greater deliberation in AI-mediated revisions.

This integrated interpretation provided a multi-level understanding of strategic regulation, linking affective triggers, cognitive-ethical regulation, and students' reported regulatory engagement. However, this integration remains interpretive in nature and does not imply that all elements of the broader conceptual framework were directly operationalized as standalone variables.

4. Results

This section presents the findings in alignment with the revised three research questions. Quantitative and qualitative results are reported in an integrated manner to document patterns of students' engagement with generative AI across regulatory behaviors, affective responses, and ethical considerations. Specifically, the section reports observed associations between AI anxiety and regulatory engagement and describes regulatory behaviors enacted during AI-assisted academic tasks.

4.1. RQ1: AI anxiety and regulatory engagement

4.1.1. RQ1. How is AI anxiety associated with university students' regulatory engagement with generative AI in academic writing tasks?

To establish continuity with prior AI anxiety research, analyses first examined the relationship between AI anxiety and **regulatory engagement** without introducing additional predictors. This step mirrors the dominant analytic structure in prior studies that model AI anxiety as a predictor of learning-related outcomes and engagement tendencies (Wang et al., 2024; Wang & Wang, 2022). In this baseline model, AI anxiety showed a statistically significant association with regulatory engagement, indicating that anxiety remains a salient affective factor in students' AI-supported learning. Table 7 reports the correlations between AI anxiety and regulatory behaviors.

The correlation analyses revealed that higher levels of AI anxiety were associated with higher frequencies of verification, revision, and selective integration, and lower levels of uncritical reliance. These bivariate patterns are summarized in Fig. 2 (Panel A), facilitating comparison across the four outcomes.

To further specify this relationship, a hierarchical regression analysis was conducted. In Step 1, AI anxiety was entered as the sole predictor of regulatory engagement, consistent with prior modeling strategies that treat anxiety as a key explanatory variable (Wang et al., 2024; Wang & Wang, 2022). Table 8 presents the hierarchical regression results

Table 6

Typology and distribution of strategic regulation in generative AI use ($N = 107$).

Regulation Type	Core Regulatory Actions	Orientation toward AI use	N	Frequency (%)
Uncritical Reliance	Direct adoption, minimal revision	AI as authority	20	18.7
Selective Integration	Partial uptake, surface revision	AI as assistant	37	34.6
Evaluative Transformation	Verification, rephrasing, synthesis	AI as provisional resource	34	31.8
Strategic Rejection	Deliberate non-use after evaluation	AI as a source of uncertainty	16	14.9

Note. Categories were developed through iterative qualitative content analysis of students' post-task reflections. Frequencies indicate the number of reflections primarily classified into each category.

Table 7

Correlations between AI anxiety and regulatory behaviors.

Variable	Verification	Revision	Selective Integration	Uncritical Reliance
AI Anxiety	.42**	.38**	.31**	-.27*

* $p < .05$, ** $p < .01$.

predicting regulatory engagement.

In line with prior studies, AI anxiety alone accounted for a significant portion of variance in regulatory engagement. However, when evaluative capacity and ethical awareness were introduced, the explanatory power of AI anxiety was substantially attenuated. To further interpret these statistical associations, the subsequent qualitative analysis examined students' reflective accounts with attention to regulatory actions, typological patterns of AI use, and the epistemic stance implied in their descriptions of questioning, revising, or accepting AI-generated content.

4.1.2. Extract 5. anxiety-triggered regulation

"Because I was worried about plagiarism, I kept questioning every sentence the AI gave me. Even when the wording sounded smooth and useful, I was not sure whether I could use it as it was, so I kept checking what needed to be changed, rewritten, or left out. That made me read the output more carefully instead of simply accepting it."

In this account, the participant described repeated questioning of AI-generated content during task completion.

4.1.3. Extract 6. ethical awareness in regulation

"Using AI made me more careful about whether this was really my work. I found myself thinking not only about whether the answer was helpful, but also about whether the final version still reflected my own ideas and effort. Because of that, I revised more carefully and paid closer attention to whether the final version still felt like my own work."

In this account, the participant described increased caution regarding authorship during AI-assisted task completion.

To further examine the conditional nature of the relationship between AI anxiety and regulatory engagement, a first-stage moderated mediation model (Hayes' PROCESS Model 7) was estimated. The AI anxiety $\times$ ethical awareness interaction significantly predicted evaluative capacity ($b = .18$, $SE = .07$, $p < .05$). The conditional indirect effect

Table 8

Hierarchical regression predicting regulatory engagement.

Predictor	β	ΔR^2
Step 1: AI Anxiety	.21*	.07
Step 2: Evaluative Capacity	.46***	.29
Step 3: Ethical Awareness	.34***	.18

* $p < .05$, *** $p < .001$.

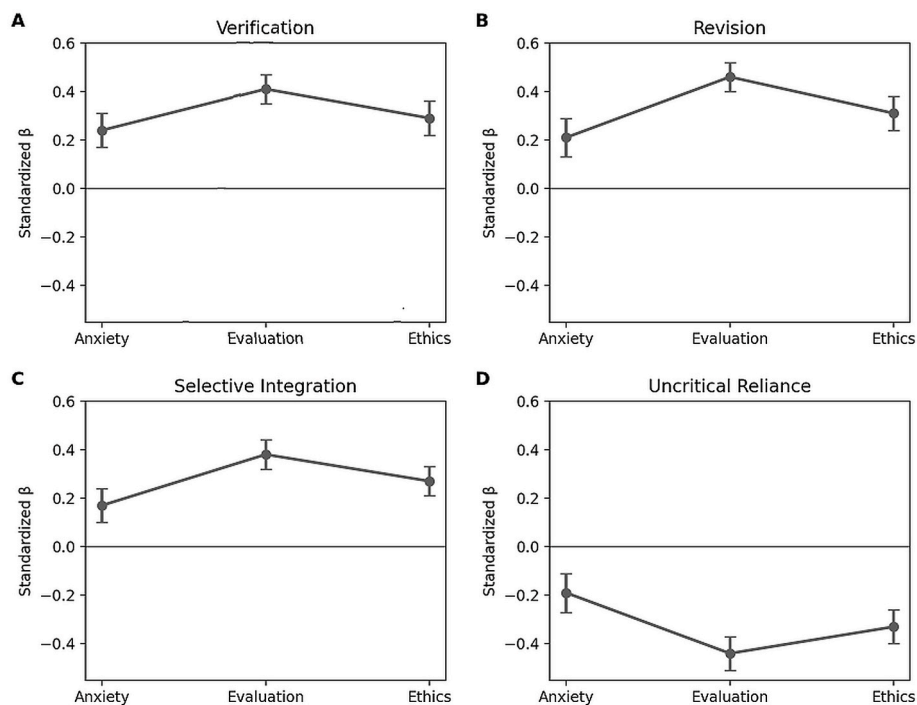


Fig. 2. Standardized coefficients (β) across four outcomes (error bars = SE).

Note. Lines show β for three predictors (AI anxiety, evaluative capacity, ethical awareness).

of AI anxiety on regulatory engagement through evaluative capacity was stronger at higher levels of ethical awareness (indirect effect = .11, 95% *CI* [.04, .21]) than at lower levels (indirect effect = .02, 95% *CI* [−.01, .08]). The index of moderated mediation was .05, with a bootstrap 95% *CI* of [.01, .12], indicating that the indirect effect varied as a function of ethical awareness. Together, these results suggest that ethical awareness may strengthen the indirect pathway from anxiety to engagement via evaluative capacity, such that anxiety may function as a more productive regulatory signal when accompanied by stronger ethical reflection.

4.2. RQ2: enacted patterns of strategic regulation

4.2.1. RQ2. How do university students describe regulating their use of generative AI through verification, revision, selective uptake, and rejection during academic writing tasks?

Qualitative analysis of students' reflective task accounts revealed that engagement with generative AI was not uniform but clustered into four distinct patterns of strategic regulation. These patterns differed in how students described their use of AI during task execution, including whether they adopted AI output directly, revised it selectively, verified it against other sources, or rejected it altogether. They also reflected different epistemic stances toward AI-generated content, ranging from uncritical acceptance to cautious scrutiny and deliberate refusal. Across the reflective accounts, selective integration and evaluative transformation appeared more frequently than either uncritical reliance or strategic rejection, suggesting that students more often regulated AI use through scrutiny and modification than through complete refusal. At the level of recurrent regulatory actions, verification and revision were more commonly described than outright rejection, indicating that students typically managed AI use through selective checking and modification rather than simple acceptance or refusal. Table 6 presents the typology of strategic regulation in students' generative AI use.

Among these categories, Evaluative Transformation was one of the most frequently observed patterns. This pattern was characterized by repeated acts of verification, reinterpretation, and selective incorporation of AI-generated content during task execution. Students in this category typically described AI as useful but not fully trustworthy, and they reported reworking AI-generated material in ways that preserved their own judgment, stance, and authorship.

4.2.2. Extract 1. evaluative transformation

"I didn't trust the explanation right away. The task was to summarize a research article, and the AI gave me something that looked too fluent to be reliable. I checked two other sources to confirm the accuracy, rewrote the idea in my own words, and only kept the structure suggested by AI."

In this account, the participant described applying verification strategies when completing a summarization task that required conceptual accuracy. The reflection shows how concern about the reliability of AI output initiated a process of cross-referencing and reinterpretation, resulting in evaluative transformation rather than direct adoption. The excerpt also illustrates how verification and rewriting were linked, suggesting that students' regulation of AI use often involved both factual checking and rhetorical reshaping.

4.2.3. Extract 2. strategic rejection

"The answer sounded confident, but I couldn't verify where it came from. I decided not to use it at all."

In this account, the participant reported deliberate non-use of AI-generated content following difficulties in source verification. This type of response was less frequent than selective integration or evaluative transformation, but it is important because it shows that strategic regulation sometimes took the form of refusal rather than revision.

4.2.4. Extract 3. selective integration

"I used the AI mostly to get ideas for how to organize my paragraph, but I did not copy the sentences directly. Some parts were useful for structure, but I changed the wording and removed examples that did not fit what I wanted to say."

In this account, the participant did not reject AI output altogether, but neither was it accepted wholesale. Instead, the reflection illustrates selective integration, in which AI functioned as a provisional support for structuring ideas while the student retained control over wording, content selection, and final meaning.

4.2.5. Extract 4. uncritical reliance

"I was short on time, so I used most of what the AI gave me. I changed a few words, but I did not really check whether the content was fully accurate. I was more focused on completing the task quickly than on checking the information carefully."

This excerpt illustrates uncritical reliance, in which AI output was adopted with minimal checking or revision. Including this pattern is important because it shows that strategic regulation was not universal across the dataset and that some students engaged with AI in more passive ways, particularly under time pressure.

Taken together, these excerpts show that students' regulatory engagement with generative AI ranged from direct adoption to deliberate rejection, with the most common patterns involving selective uptake, revision, and verification rather than passive acceptance.

4.3. RQ3: evaluative judgment, ethical awareness, and strategic AI use

4.3.1. RQ3. How do students' reflective accounts illuminate the roles of evaluative judgment and ethical awareness in strategic AI use?

Analyses associated with RQ3 examined mean differences in AI use outcomes across three levels of evaluative capacity. Table 9 reports group means (*SD*) and one-way ANOVA results for surface use, transformative use, and learning ownership. These analyses were used to support interpretation of how evaluative capacity was associated with different patterns of AI-supported learning, rather than to claim that all outcomes represented directly measured components of a single validated causal model.

Students with higher evaluative capacity reported lower surface use and higher transformative use and learning ownership. One-way ANOVA indicated significant group differences for surface use, transformative use, and learning ownership (Table 9; $F(2,104) = 5.40$ – 8.07 , $p \leq .006$, $\eta^2 = .09$ – $.13$), and Tukey HSD tests showed that the Low group differed most consistently from the High (and Medium) group across outcomes. Taken together, these patterns suggest that stronger evaluative capacity was associated with more active and reflective forms

Table 9

Mean (*SD*) differences in AI use outcomes by evaluative capacity level ($n = 107$).

Evaluative capacity level	<i>n</i>	Surface use <i>M</i> (<i>SD</i>)	Transformative use <i>M</i> (<i>SD</i>)	Learning ownership <i>M</i> (<i>SD</i>)
Low	50	2.41 (.89)	3.88 (.67)	3.47 (.75)
Medium	28	2.27 (.77)	4.35 (.59)	3.96 (.65)
High	29	1.78 (.67)	4.33 (.91)	4.09 (.78)
ANOVA		$F(2,104) = 5.69$, $p = .005$, $\eta^2 = .10$	$F(2,104) = 5.40$, $p = .006$, $\eta^2 = .09$	$F(2,104) = 8.07$, $p < .001$, $\eta^2 = .13$
Post-hoc (Tukey HSD)		Low > High ($p = .003$)	Low < Medium ($p = .019$); Low < High ($p = .024$)	Low < Medium ($p = .013$); Low < High ($p = .001$)

Note. Evaluative capacity levels were created using tertiles. Group differences were tested using one-way ANOVA with Tukey HSD post-hoc comparisons; η^2 is reported as the effect size.

of AI-supported learning.

4.3.2. Extract 7. transformation through regulation

“AI helped me think, but the final idea felt more mine because I changed it so much. It was for a short essay on ethical dilemmas, and I realized that the AI’s argument didn’t fully match my stance. I used it to outline my reasoning but rewrote all the supporting examples.”

This reflection illustrates a more active and reflective mode of AI-supported learning, where students not only revised AI-generated text but also reasserted authorship and evaluative control during the task. It also suggests that reflective AI use involved not merely correcting language, but reshaping argument, stance, and ownership in relation to the student’s own intentions.

4.3.3. Extract 8. ownership through regulation

“I didn’t feel like I was cheating. I felt like I was learning how to think better, because I was not simply copying what the AI gave me. I kept changing the wording, rethinking the ideas, and deciding what really fit my argument. In that process, I felt that I was not just using AI to finish the task more quickly, but learning how to think more carefully and make the final work my own.”

This comment complements the previous reflection, emphasizing a sense of ownership and ethical responsibility as features of students’ reported regulatory engagement. Together, these excerpts suggest that students’ more reflective accounts of AI use were closely tied to judgments about authorship, accountability, and whether the final text could still be regarded as their own work.

Table 10 summarizes the hierarchical regression results ($N = 107$), including the interaction between evaluative capacity and ethical awareness. Fig. 2 visualizes standardized coefficients (β) and SE across outcomes, and Fig. 3 contrasts bivariate associations with multivariate effects. As shown in Table 10, evaluative capacity consistently emerged

Table 10

Effects of AI anxiety, evaluative capacity, and ethical awareness on regulatory engagement and interaction effects ($N = 107$).

Outcome Variable	Predictor	β	SE	t	p	ΔR^2
Verification	AI Anxiety	.24	.07	3.42	.001	.06
	Evaluative Capacity	.41	.06	6.83	<.001	.18
	Ethical Awareness	.29	.07	4.14	<.001	.11
	AI Anxiety	.21	.08	2.76	.007	.05
Revision	Evaluative Capacity	.46	.06	7.58	<.001	.21
	Ethical Awareness	.31	.07	4.43	<.001	.12
	AI Anxiety	.17	.07	2.31	.023	.04
	Evaluative Capacity	.38	.06	6.12	<.001	.16
Selective Integration	Ethical Awareness	.27	.06	4.02	<.001	.10
	AI Anxiety	-.19	.08	-2.48	.015	.04
	Evaluative Capacity	-.44	.07	-6.29	<.001	.19
	Ethical Awareness	-.33	.07	-4.71	<.001	.13
Uncritical Reliance	Evaluative Capacity	.41**	—	—	<.01	—
	Ethical Awareness	.47**	—	—	<.01	—
	Evaluative Capacity × Ethical Awareness	—	—	—	<.01	—
	Evaluative Capacity × Ethical Awareness	—	—	—	<.01	—

Note. Standardized coefficients (β) are reported. ΔR^2 indicates incremental variance explained by each predictor block. * $p < .05$, ** $p < .01$, *** $p < .001$.

as the strongest predictor across regulatory engagement outcomes, indicating that AI-supported learning is shaped less by affective dispositions than by learners’ evaluative regulation. Ethical awareness demonstrated consistent positive associations with evaluative regulatory behaviors and a negative association with uncritical reliance. AI anxiety showed smaller but statistically significant associations across outcomes, with positive coefficients for verification, revision, and selective integration, and a negative coefficient for uncritical reliance. These findings support the interpretation that evaluative judgment and ethical awareness are important in explaining how students move from simple AI use toward more selective, reflective, and responsible engagement.

Fig. 2 visualizes the standardized regression coefficients reported in Table 10 across the four outcomes. The plots highlight a consistent pattern: coefficients are positive for verification, revision, and selective integration, whereas all predictors show negative coefficients for uncritical reliance.

Fig. 3 provides a compact comparison between the bivariate associations and the multivariate effects across the four outcomes. Panel A summarizes the zero-order correlations (r) between AI anxiety and each outcome, whereas Panels B–D display standardized regression coefficients (β) with SE error bars from the multivariate models. Overall, the pattern is consistent in direction: AI anxiety, evaluative capacity, and ethical awareness show positive associations with verification, revision, and selective integration, but negative associations with uncritical reliance. This visual comparison clarifies which relationships remain robust when predictors are considered simultaneously.

4.4. Integrated interpretation of quantitative and qualitative findings

The integration of quantitative and qualitative findings revealed a multi-layered pattern of regulatory engagement through which affective, cognitive, and ethical dimensions jointly shape students’ engagement with generative AI. As shown in Fig. 4, the affective layer (AI anxiety) may be interpreted as one important condition associated with closer scrutiny of AI outputs, while evaluative capacity and ethical awareness help explain how students move toward more selective, reflective, and responsible AI use. Rather than presenting a validated causal sequence, this integrated interpretation is intended as a heuristic account of how the measured variables and qualitative insights can be read together.

This integrated account synthesizes statistical associations, qualitative insights, and theoretical interpretation into a heuristic interpretive framework, linking affective, evaluative, and ethical dimensions of regulation to students’ reported AI-supported learning outcomes. Not all elements shown in this integrated account were directly measured as standalone variables; some remain interpretive concepts used to explain recurring patterns in the data.

5. Discussion: from dispositional predictors to strategic regulation

Fig. 4 consolidates the integrated findings, positioning AI literacy as an emergent competence arising from the interplay of affective, cognitive, and ethical regulation. This figure synthesizes the quantitative and qualitative results presented earlier (Table 5), providing a visual summary of the interpretive links among affective-cognitive-ethical dimensions of regulation in AI-supported learning.

This study advances research on generative artificial intelligence (AI) in higher education by reconfiguring how AI anxiety, attitudes, and learning engagement are analytically positioned. Rather than contesting existing findings, the present study extends prior work by shifting the level of analysis from dispositional predictors of AI use to the regulatory practices through which learning with AI is enacted. Across the three research questions, the results document patterned associations among AI anxiety, regulatory behaviors, evaluative capacity, and ethical

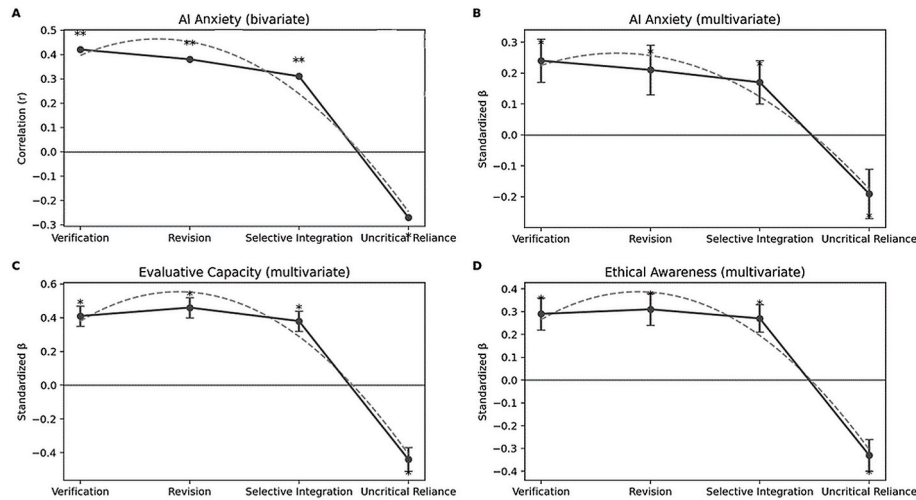


Fig. 3. Bivariate and multivariate effects of AI anxiety, evaluative capacity, and ethical awareness across four regulatory outcomes (error bars = SE). **Note.** Panel A shows bivariate correlations (r), and Panels B–D present standardized regression coefficients (β) with standard errors, highlighting consistent relationships when predictors are modeled together.

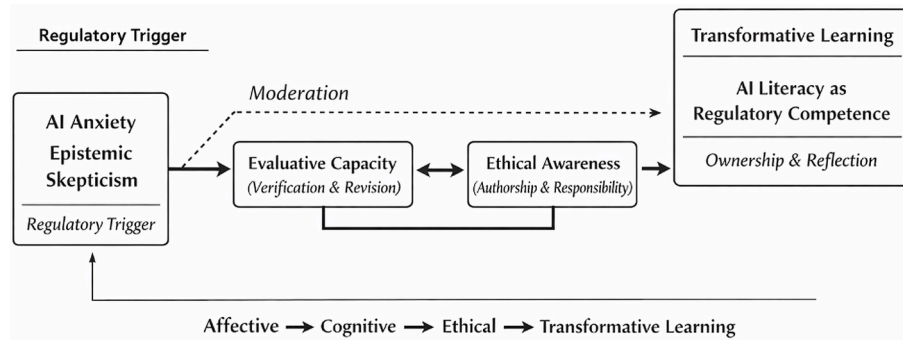


Fig. 4. Integrated Framework of Affective-Cognitive-Ethical Regulation in AI-mediated Learning. **Note.** The figure should be interpreted as a heuristic representation rather than a validated causal model.

awareness in students' AI-assisted academic work, providing an integrated account of how AI engagement unfolds in practice.

Prior studies have convincingly demonstrated that AI anxiety and attitudes are salient affective-dispositional factors shaping learners' receptivity to AI technologies and their likelihood of adoption or use (Davis, 1989; Kaya et al., 2024; Schepman & Rodway, 2023; Wang et al., 2024; Wang & Wang, 2022). These contributions have been foundational in explaining why learners differ in their orientations toward AI and in establishing robust predictive relationships among psychological variables. The present findings do not challenge these insights. Rather, they extend them by suggesting that much of the recent GenAI-in-higher-education literature, including current work on AI anxiety, trust, and AI literacy, has primarily operated at an analytical layer concerned with adoption, intention, and general engagement tendencies, whereas the present study focuses more explicitly on how learners regulate AI use during actual academic work. This contrast is consistent with recent review-based and higher-education-focused scholarship showing that generative AI in education must be examined not only in terms of uptake or engagement, but also in terms of pedagogical design, responsible use, and the conditions under which learners exercise judgment over AI-generated content (Chiu, 2024; Lee et al., 2024; Lo et al., 2024). In this respect, the findings are more closely aligned with recent discussions of evaluative judgment, responsible AI use, and process-oriented AI literacy in higher education than with models centered primarily on acceptance or frequency of use.

By contrast, the current study examines how learning with

generative AI is enacted at a regulatory layer. By first establishing continuity with prior AI anxiety research through a baseline model and then demonstrating how anxiety's explanatory role is reshaped once evaluative capacity and ethical awareness are considered, the findings position AI anxiety not as a direct determinant of learning outcomes but as one component within a broader regulatory configuration. This shift from predictive to explanatory analysis enables a more fine-grained understanding of how AI becomes a learning resource in academic contexts.

5.1. AI anxiety revisited: from dispositional predictor to regulatory signal

Regression results (Table 10) indicate that AI anxiety plays a conditional rather than primary role in AI-supported learning (see Figs. 2 and 4). While anxiety remained significant, its explanatory power diminished once evaluative capacity and ethical awareness were included, showing that learners' regulatory resources shape engagement more directly. The moderated mediation analysis (PROCESS Model 7) confirmed that anxiety's indirect effect on engagement through evaluative capacity was stronger at higher levels of ethical awareness, suggesting that anxiety may contribute to more productive engagement when paired with ethical reflection.

Thus, AI anxiety may be interpreted as an affective signal associated with closer scrutiny and more cautious AI-supported academic work. Rather than functioning solely as a dispositional constraint, it may, under certain conditions, contribute to more deliberate regulatory

engagement. Differences in previous findings—where anxiety appeared strong, weak, or nonsignificant (Kaya et al., 2024; Wang et al., 2024; Wang & Wang, 2022)—likely reflect variation in learners' regulatory competence rather than inconsistencies in the construct itself.

5.2. Attitudes and intentions as pre-learning conditions

The findings also clarify the explanatory boundaries of attitude- and intention-based models of AI use. Research employing general attitude measures toward AI has demonstrated systematic associations with personality traits, institutional trust, and demographic variables (Schepman & Rodway, 2023; Kaya et al., 2024), offering valuable insight into why learners differ in their willingness to engage with AI technologies.

However, the present results suggest that such constructs operate primarily as pre-learning conditions rather than as explanations of learning itself. Students with comparable attitudes toward AI enacted markedly different regulatory practices during AI-assisted tasks. These differences cannot be fully accounted for by attitudes or intentions alone, lending empirical support to critiques that adoption-centered models conflate technology use with meaningful learning engagement (Davis, 1989; Selwyn, 2019).

In the context of generative AI—where outputs may be fluent yet epistemically unstable or ethically ambiguous—the central pedagogical issue shifts from whether students intend to use AI to how they evaluate, modify, or reject AI-generated content. Attitudes and intentions may predict frequency of use, but they offer limited insight into the regulatory work through which AI outputs are transformed into knowledge. By foregrounding enacted practices, the present study complements dispositional accounts with a process-oriented explanation of learning.

5.3. Strategic regulation as an interpretive framework

Strategic regulation provides an interpretive framework through which these enacted practices can be understood. The qualitative analyses revealed patterned regulatory actions—verification, rephrasing, synthesis, and strategic rejection—that reflect learners' active negotiation of AI's epistemic role during task execution. These findings extend self-regulated learning research by situating regulation not only within learners' internal cognition but within the learner-AI epistemic relationship (Bandura, 2002; Piniel & Csizér, 2013; Zimmerman, 2000).

The interaction between evaluative capacity and ethical awareness is particularly instructive. Prior studies have examined critical thinking, ethical concern, or self-efficacy in isolation (Ng et al., 2021; Wang et al., 2024), correctly identifying important dimensions of AI engagement. The present findings suggest that more reflective and responsible AI-supported learning may be strengthened when these two dimensions operate together. Evaluative capacity without ethical awareness risks instrumental or opportunistic AI use, whereas ethical concern without evaluative skill may lead to avoidance. Strategic regulation arises when learners are able to evaluate AI outputs while simultaneously negotiating responsibility and authorship.

The significant interaction effects reported in Table 10 further indicate that evaluative capacity alone is insufficient to account for more active forms of engagement with AI. Ethical awareness appears to amplify the impact of evaluative regulation, enabling learners to negotiate responsibility and authorship when working with AI-generated content.

This interaction helps explain why generative AI does not automatically function as a learning resource. As critical scholars have argued, AI systems participate in meaning-making and thereby reshape epistemic relations in education (Johnson & Verdicchio, 2017; Selwyn, 2019). The present study suggests that more meaningful AI-supported learning is more likely when learners actively regulate this participation. Strategic regulation, therefore, is not simply an ancillary skill; rather, it is a useful interpretive framework for understanding how

students maintain judgment and responsibility in AI-mediated learning. This suggests that AI literacy may be more productively understood not as tool acceptance alone, but as regulatory competence grounded in evaluative judgment, ethical awareness, and responsible academic decision-making.

5.4. From acceptance to enactment

Taken together, the findings support a shift from acceptance-based to enactment-based models of AI-supported learning. Dominant frameworks have rightly focused on whether learners accept AI or intend to use it (Davis, 1989; Wang & Wang, 2022). The present study extends this work by examining how AI use is enacted once adoption has occurred. This shift aligns with broader movements in educational technology research that emphasize agency, practice, and context over adoption metrics (Ng et al., 2021; Selwyn, 2019).

By integrating affective (AI anxiety), cognitive (evaluative capacity), and ethical (ethical awareness) dimensions within a regulatory framework, this study bridges dispositional predictors and learning processes. Anxiety and attitudes remain important, but their educational significance becomes intelligible only when situated within the regulatory practices through which learners manage epistemic risk. At the same time, these implications should be interpreted in light of the bounded context of the study, namely a South Korean university setting, relatively short English-writing tasks, and post-task reflections based on a single AI-use episode.

5.5. Implications for research and pedagogy

For research, the findings highlight the need to move beyond designs relying solely on attitudes or intentions as outcome variables. Future studies should integrate dispositional measures with analyses of enacted regulation, employing mixed-methods approaches that capture how learners actually work with AI during authentic academic tasks (Creswell & Plano Clark, 2018).

For pedagogy, the results indicate that reducing AI anxiety should not be the primary instructional goal. Instead, instruction should cultivate evaluative capacity and ethical awareness, enabling students to regulate their engagement with AI effectively. In this respect, the findings align with recent discussions of AI literacy pedagogy that emphasize evaluative judgment, responsible authorship, guided reflection, and the pedagogical conditions that support meaningful engagement with generative AI rather than mere tool familiarity or usage frequency (Chiu, 2024; Lee et al., 2024; Lo et al., 2024). Rather than teaching AI as a technical tool, educators can embed reflection-based strategies that link evaluation and ethics. One possible way to operationalize this is through an AI-Regulation Reflection Protocol (AIRP) that can be incorporated into coursework:

- (1) students use AI to complete academic tasks and assess its outputs for credibility and coherence through source verification;
- (2) they revise AI-generated wording while documenting key changes through simple revision logging;
- (3) they explicitly consider attribution and authorship responsibility when deciding what can be retained;
- (4) they record brief reflections explaining why they accepted, modified, or rejected AI suggestions; and
- (5) they engage in guided reflection and prompt critique with peers or instructors to strengthen evaluative and ethical reasoning.

This iterative reflection process can position AI use as a more metacognitive and ethically grounded practice, supporting a view of AI literacy as regulatory competence, namely the capacity to exercise evaluative judgment and responsible decision-making in AI-mediated learning (Ng et al., 2021; Selwyn, 2019). Although this pedagogical protocol was not itself tested as an intervention in the present study, it is

offered as a practice-oriented implication derived from the overall pattern of findings.

5.6. Pedagogical implications and model for AI literacy as regulatory competence

The findings of this study suggest a pedagogical reorientation in AI literacy education, shifting from procedural training on tool use toward cultivating learners' regulatory competence in evaluative and ethical judgment during AI-supported academic work. Building on the framework of strategic regulation, a Pedagogical Model for AI Literacy through Regulation is proposed to translate the study's theoretical insights into instructional practice. This model is presented as a practice-oriented implication of the findings rather than as an instructional intervention directly tested in the present study.

5.6.1. Affective-cognitive awareness phase: from AI-related concern to evaluative attention

In the initial phase, instruction should help students recognize concern or uncertainty in AI use not as fear of AI but as a potential starting point for critical engagement. Classroom activities may include guided evaluations of AI outputs, structured error analysis, and reflection on epistemic uncertainty. The goal is to normalize evaluative attention—teaching students how to question, verify, and contextualize AI-generated knowledge rather than merely accept or reject it.

5.6.2. Ethical-reflective regulation phase: from monitoring to responsible use

The second phase focuses on developing ethical awareness in practice by embedding ethical reasoning into academic task design. Students can be encouraged to document their AI interaction processes—detailing how they revised, attributed, or rejected AI suggestions—and to discuss issues of authorship, originality, and responsibility. Through these reflective protocols, ethical awareness becomes an active regulatory practice rather than a declarative principle.

5.6.3. Integrative application phase: from regulation to more reflective and responsible AI use

In the final phase, learners synthesize evaluative and ethical regulation into more reflective and responsible forms of AI-supported learning. Project-based tasks can be structured where students iteratively use AI tools while maintaining authorship control and evaluative responsibility. Educators act as facilitators who model reflective verification and help learners articulate why and how they rely on AI, transforming regulation into more deliberate academic decision-making.

5.6.3.1. Implementation considerations. Curricular Integration. The model can be incorporated into writing-intensive or inquiry-based courses where generative AI use is optional but scaffolded through reflective checkpoints.

Instructor Role. Teachers function as regulation coaches, supporting metacognitive questioning, monitoring students' evaluative strategies, and modeling ethical deliberation.

Assessment. Evaluation should move beyond product quality to include process transparency, assessing how students document and justify their regulatory decisions.

5.6.3.2. Educational implications. This model reframes AI literacy not as knowing how to operate AI but as knowing how to regulate learning with AI. It positions evaluation and ethical awareness as dual competencies essential for responsible AI engagement in higher education. By operationalizing strategic regulation in pedagogy, educators can approach generative AI not only as a potential epistemic risk but also as a context for cultivating reflective, autonomous, and ethically grounded

learning.

5.6.3.2.1. Summary of contributions. This study makes contributions at three interrelated levels, theoretical, empirical, and pedagogical, that collectively advance understanding of AI-mediated learning:

- **Theoretical Contribution:** The study advances a conceptual shift from adoption-based to enactment-based models of AI-supported learning, reconfiguring AI anxiety from a dispositional constraint to a potential regulatory signal within a broader affective–evaluative–ethical configuration.
- **Empirical Contribution:** Through mixed-methods analysis, the study identifies distinct regulatory patterns (uncritical reliance, selective integration, evaluative transformation, strategic rejection) and clarifies how evaluative and ethical resources interact to shape students' reported regulatory engagement and more reflective AI-supported learning practices.
- **Pedagogical Contribution:** The findings translate into a pedagogical framework for AI literacy as regulatory competence, emphasizing evaluative judgment, ethical and authorship-aware AI use, instructional scaffolding, and reflective engagement as core instructional objectives in higher education.

Together, these contributions establish a coherent foundation for future research and pedagogy aimed at cultivating learners' **judgment, responsibility, and reflective engagement** in AI-mediated learning environments.

5.7. Limitations and future research

This study has several limitations. First, the sample was drawn from a single general education English course in one institutional context, which may limit the transferability of the findings to other disciplines, levels, or instructional designs. More specifically, the study was conducted in a South Korean university context and focused on relatively short English-writing tasks, so the findings should be interpreted with caution beyond similar curricular and linguistic settings. Second, key constructs and AI-use outcomes were measured primarily through self-report surveys and brief post-task reflections based on a single AI-use episode; thus, the results may be influenced by recall bias and social desirability and may not capture longer-term patterns of AI-supported learning. Third, the design was cross-sectional, and the analyses capture conditional associations rather than causal effects. Fourth, regulatory engagement was operationalized through verification, revision, selective integration, and reduced uncritical reliance rather than direct performance gains; future work should incorporate task artifacts, external ratings, and longitudinal measures of learning outcomes. Finally, because AI use was voluntary and device-based, the specific generative AI tool versions and interfaces could vary across students, and the study therefore emphasizes regulation processes over tool-specific comparisons. Future research should triangulate self-report with trace data (e.g., interaction logs), replicate the model across task types, and test instructional interventions that scaffold evaluative and ethical regulation in AI-mediated learning.

6. Conclusion: from adoption to enactment in AI-supported learning

This study set out to move beyond adoption-centered explanations of generative AI use in higher education by examining how students enact strategic regulation in AI-mediated learning contexts. Drawing on mixed-methods evidence, the findings extend prior research on AI anxiety and attitudes by situating these constructs within a process-oriented account of learning. Rather than rejecting existing models, the study suggests that their explanatory power is bounded by the analytical level at which AI engagement has typically been examined.

Quantitatively, AI anxiety modestly predicted engagement ($\beta = .21$,

$\Delta R^2 = .07$), but the inclusion of evaluative capacity and ethical awareness substantially increased the explained variance ($\Delta R^2 = .47$), showing that regulatory competence was more strongly associated with meaningful AI-supported learning. Qualitative findings complemented this pattern, illustrating how learners verified, revised, and ethically negotiated AI-generated content to transform uncertainty into more deliberate and responsible engagement.

First, the findings indicate that AI anxiety cannot be adequately conceptualized as a unidimensional barrier to learning. Consistent with distinctions in educational psychology between facilitating and debilitating anxiety (Alpert & Haber, 1960; Piniel & Csizér, 2013), AI anxiety emerged as a regulatory signal whose educational consequences depend on learners' evaluative and ethical resources. This perspective helps reconcile inconsistencies in prior studies, where AI anxiety has alternately appeared as a strong negative predictor, a weak correlate, or a non-significant factor (Kaya et al., 2024; Wang et al., 2024; Wang & Wang, 2022).

Second, the study clarifies the limits of attitude- and intention-based models of AI use. While such models have been effective in explaining acceptance and frequency of use (Davis, 1989; Schepman & Rodway, 2023), they operate at a level that precedes learning itself. The present findings show that learning outcomes are shaped less by whether students intend to use AI than by how they evaluate, modify, and ethically negotiate AI-generated content during task performance. This distinction is particularly consequential in the context of generative AI, where fluent outputs can obscure epistemic risk if left unregulated (Mavroudi et al., 2025; Ng et al., 2021; Selwyn, 2019).

Third, by introducing the concept of strategic regulation, this study provides a theoretical framework for interpreting how generative AI may function as a learning resource in academic contexts. Strategic regulation captures the interplay of evaluative judgment, ethical awareness, authorship-sensitive AI use, and instructional scaffolding through which learners negotiate AI's epistemic authority. The observed interaction between evaluative capacity and ethical awareness underscores that productive AI-supported learning is neither purely cognitive nor purely moral, but inherently regulatory. In this respect, the study extends theories of self-regulated learning by relocating the object of regulation from learners' internal cognition to the learner-AI epistemic relationship (Bandura, 2002).

Taken together, these contributions support a meaningful conceptual shift from acceptance-based to enactment-based models of AI-supported learning. Rather than focusing primarily on predicting AI use, future research should examine how learning is enacted in practice and how learners assert judgment and responsibility in AI-mediated environments. In instructional terms, this points to the importance of scaffolding strategic regulation through evaluative judgment, ethical and authorship-aware AI use, and guided reflection rather than focusing only on attitudes toward AI. Ultimately, generative AI does not determine

learning outcomes by itself. Rather, the findings underscore the importance of learners' regulatory competence: their ability to evaluate, revise, justify, and take responsibility for AI-supported academic work.

Ethics statement

This study was conducted in accordance with institutional policies governing routine classroom-based educational activities. Formal ethics committee approval was not required under the author's institutional guidelines because the study involved analysis of anonymized student work and reflections produced as part of regular course activities. Informed consent was obtained from all participants prior to data collection, and all data were de-identified before analysis.

Data availability statement

The data supporting the findings of this study are not publicly available because they include de-identified student reflections produced in an educational context and are therefore subject to ethical and privacy restrictions. The data may be made available from the corresponding author upon reasonable request, subject to institutional and ethical considerations.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the author used ChatGPT and Google Translate for language support only, including grammar checking. After using these tools, the author carefully reviewed, revised, and edited the manuscript as needed and takes full responsibility for the content of the published article. No generative AI tool was used to generate, interpret, or validate the study's data or findings.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Survey Items for Key Constructs

Construct	Item Example(s)	Source/Notes
AI Anxiety	"I feel uneasy when I rely on AI-generated information for learning."	Adapted from Wang and Wang (2022)
Evaluative Capacity	1. "I check the accuracy of AI-generated information against other sources." 2. "I revise AI-generated text to fit my understanding."	Developed from Ng et al. (2021), adapted for generative AI context.
Ethical Awareness	1. "I consider whether using AI-generated content affects the originality of my work." 2. "I feel responsible for acknowledging AI assistance in my academic writing."	Adapted from Ng et al. (2021)
Regulatory Engagement	1. "When using AI, I actively reflect on how it influences my own thinking." 2. "I reject AI-generated content during academic writing tasks."	Developed for this study.

Note. All items were rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Cronbach's α coefficients for each subscale exceeded .80, indicating acceptable internal consistency.

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Eunjeon Kim is a faculty member in the Graduate School of Education at Dankook University, Republic of Korea. Her research interests include AI-assisted language learning, EFL pedagogy, learner discourse, and the analysis of instructional artifacts in technology-mediated educational contexts.